

The Hubble-Kernel Growth Equation: A Parameter-Free, Falsifiable Late-Time Growth Law from a de Sitter-Locked MOND Scale

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AI-assisted research programme; not peer reviewed

11 September 2026

Abstract

If the galactic acceleration scale is locked to the dark-energy density, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, and the MOND kernel on cosmological scales sees the Hubble-flow acceleration $cH(z)$ as its ambient field (the only prescription that survives a lensing test in the accompanying repository: a kernel acting on the peculiar field alone amplifies linear-scale potentials tenfold), then the effective gravitational coupling of linear structure is fixed with no new parameter: $G_{\text{eff}}(z)/G = \nu(cH(z)/a_0)$ with $(cH_0/a_0)^2 = 8\pi/(3\kappa^2\Omega_\Lambda) = 49$ for $\kappa = \frac{1}{2}$, i.e. $cH_0/a_0 = 7.0$, and ν the radial-acceleration-relation function already used in galaxies. It predicts $G_{\text{eff}}/G = 1.077, 1.050, 1.030, 1.004$ at $z = 0, 0.5, 1, 3$ (canonical footing), hence σ_8 raised by 1.1% (1.5% on the alternative footing) over Planck- Λ CDM, $S_8 = 0.843$ (0.847), and $f\sigma_8$ raised by 2.7%, 1.9%, 1.1% at $z = 0.3, 0.6, 1.0$ (3.8, 2.6, 1.6%). The sign is opposite to the reported S_8 tension; the amplitude is inside current redshift-space-distortion errors ($\sim 5\%$) and at the reach of DESI’s full sample (1–2%). The identity is machine-checked (Lean 4); the growth computation is a committed script. This is a prediction of the framework’s cosmology under the stated prescription, with the dark component clustering as in Λ CDM; it leaves the galaxy-scale phenomenology untouched and can be killed by a low- S_8 or low- $f\sigma_8$ result at the few-percent level.

1 Derivation

With $\rho_\Lambda = 3\Omega_\Lambda H_0^2/(8\pi G)$ and $a_0 = \kappa c \sqrt{G \rho_\Lambda}$,

$$\left(\frac{cH_0}{a_0}\right)^2 = \frac{c^2 H_0^2}{\kappa^2 c^2 G \rho_\Lambda} = \frac{8\pi}{3\kappa^2 \Omega_\Lambda}, \quad (1)$$

a pure identity (Lean theorem `hubble_kernel_identity`). With $\kappa = \frac{1}{2}$ and $\Omega_\Lambda = 0.686$ this is 49.0, so $cH_0/a_0 = 7.0$; the alternative footing ($a_0 = 1.13 \times 10^{-10} \text{ m s}^{-2}$) gives 5.8. On cosmological scales the peculiar acceleration is far below a_0 , so the kernel’s behaviour is set by its ambient argument. Two prescriptions were computed with a QUMOND particle-mesh code validated against halofit: a kernel on the peculiar field alone overproduces the lensing (potential) power at $0.15\text{--}0.5 h \text{ Mpc}^{-1}$ by $9\text{--}25\times$ at $z = 0$ and is excluded; a kernel whose argument includes $cH(z)$ is Newtonian to within $\nu(cH/a_0) - 1$. The surviving prescription therefore gives the scale-independent, redshift-dependent

coupling

$$\boxed{\frac{G_{\text{eff}}(z)}{G} = \nu\left(\frac{cH(z)}{a_0}\right) = \frac{1}{1 - \exp\left[-\sqrt{7.0} E(z)\right]}, \quad E(z) = \frac{H(z)}{H_0},} \quad (2)$$

using $\nu(x) = 1/(1 - e^{-\sqrt{x}})$, the radial-acceleration-relation function.

2 Prediction

Linear growth with $\delta'' + (2 + d \ln H / d \ln a) \delta' = \frac{3}{2} \Omega_m(a) [G_{\text{eff}}(a)/G] \delta$ from $z = 100$, Λ CDM background ($\Omega_m = 0.314$, $h = 0.674$):

	canonical ($\kappa = \frac{1}{2}$)	alternative footing
G_{eff}/G at $z = 0/0.5/1/3$	1.077/1.050/1.030/1.004	1.099/1.067/1.042/1.006
σ_8 relative to Planck- Λ CDM	+1.1%	+1.5%
S_8 (Planck- Λ CDM 0.834)	0.843	0.847
$f\sigma_8$ shift at $z = 0.3/0.6/1.0$	+2.7/ + 1.9/ + 1.1%	+3.8/ + 2.6/ + 1.6%

The sign is opposite to the S_8 tension reported by KiDS and DES. A weak-lensing S_8 two to three percent below Planck- Λ CDM, or DESI $f\sigma_8$ at $z \simeq 0.3$ – 0.6 two to three percent below Λ CDM, falsifies the equation. Scope: the framework’s cosmology under the Hubble-flow prescription, with the dark component clustering as cold matter; the galaxy-scale gates are unaffected. Provenance: repository lanes L176–L180 (`fable_independent_2026/`), Lean file `Mondlean.lean` (100 theorems, zero `sorry`); the a_0 – ρ_Λ relation and its coefficient status are in doi:10.5281/zenodo.22559892.